Angular Advanced - Firebase – Realtime databases

Peter Kassenaar –
info@kassenaar.com

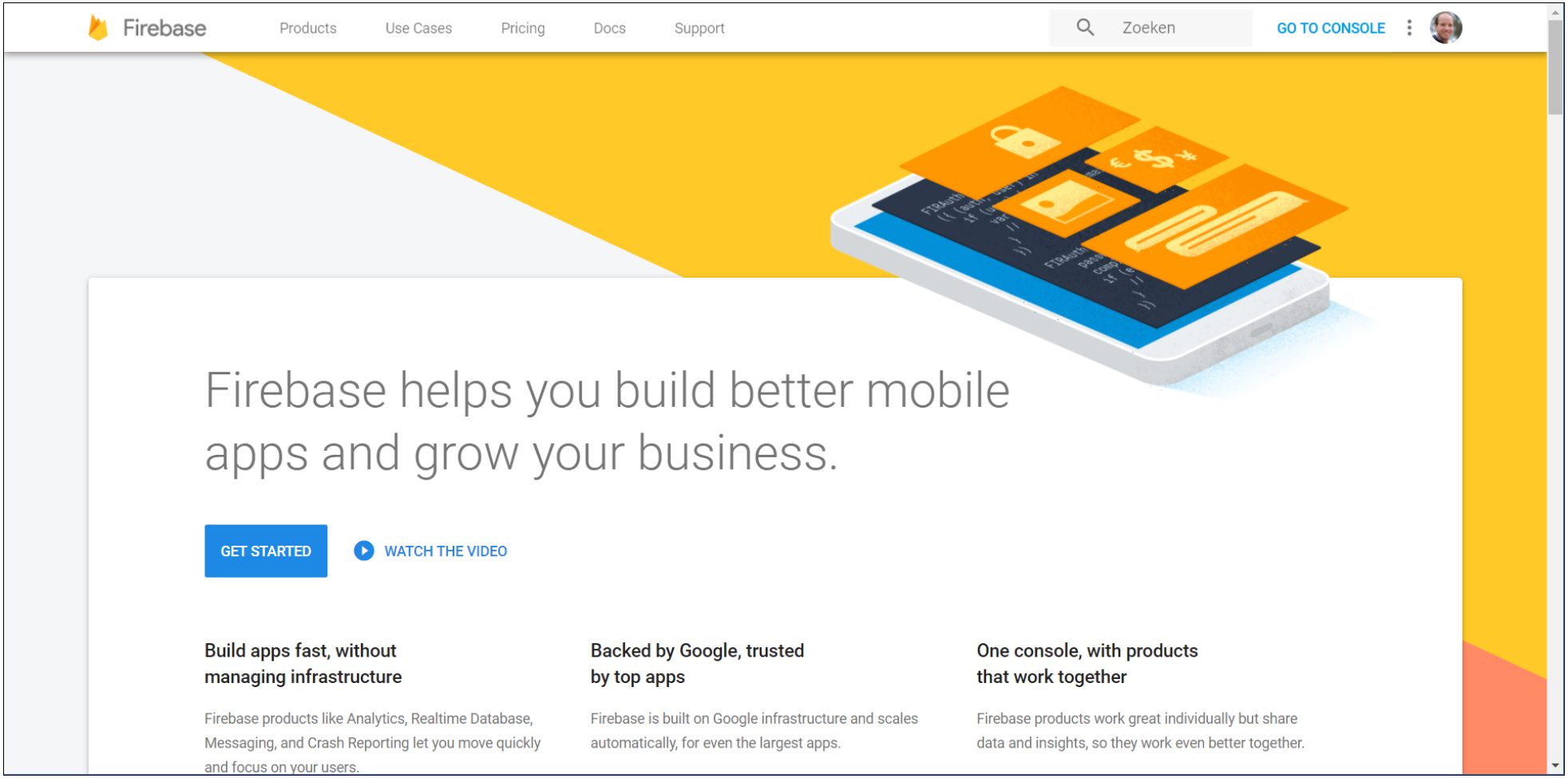
ANGULAR

Firebase

Firebase – Realtime database

- Free, online cloud database
- NoSQL database
- Feed it just JSON
- No need to setup structure or tables
- Relies *heavily* on observables
- All connected clients get latest data
- Served over websockets, `ws://`

https://firebase.google.com/



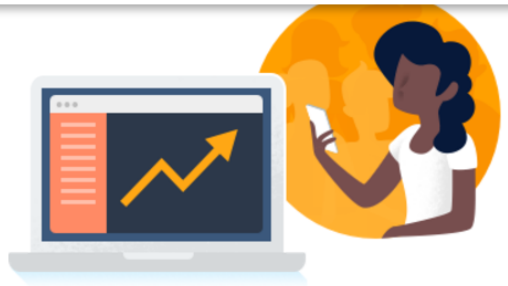
More than just a database....

Firebase

Products Use Cases Pricing Docs Support

Zoeken

GO TO CONSOLE



Develop & test your app

Realtime Database
Store and sync app data in milliseconds

Crash Reporting
Find and prioritize bugs; fix them faster

Authentication
Authenticate users simply and securely

Cloud Functions
Run mobile backend code without managing servers

Cloud Storage
Store and serve files at Google scale

Hosting
Deliver web app assets with speed and security

Test Lab for Android
Test your app on devices hosted by Google

Performance Monitoring
Gain insight into your app's performance

Grow & engage your audience

Google Analytics
Get free and unlimited app analytics

Cloud Messaging
Send targeted messages and notifications

Dynamic Links
Drive growth by using deep links with attribution

Remote Config
Modify your app without deploying a new version

Invites
Make it easy to share your app and content

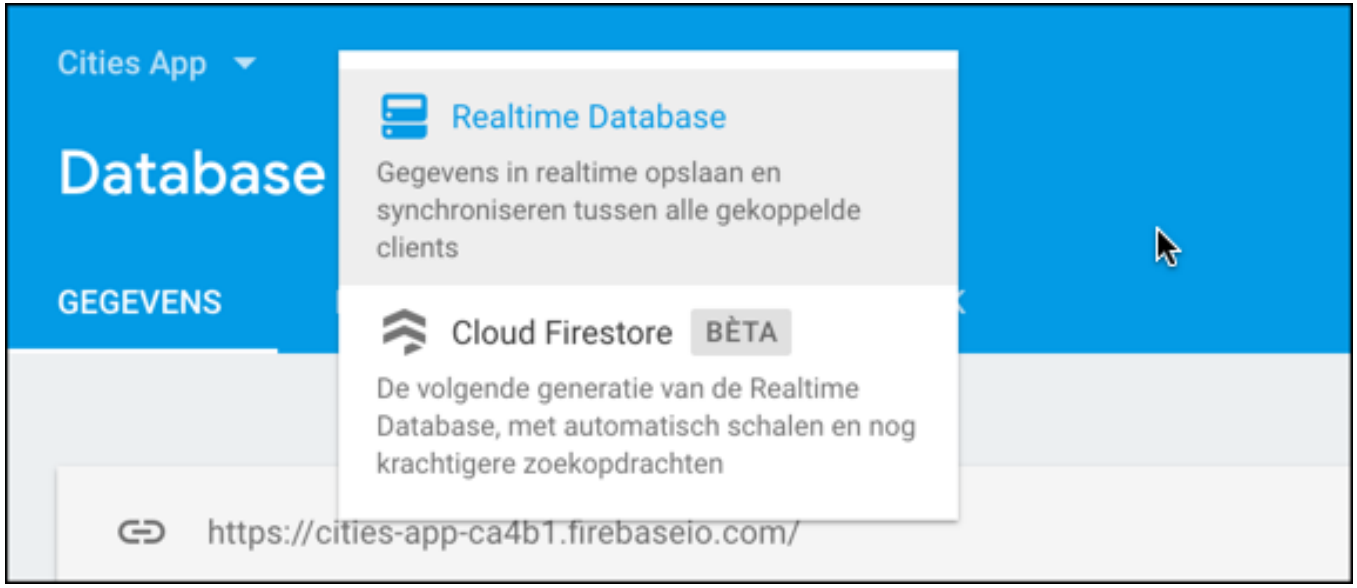
App Indexing
Drive search traffic to your mobile app

AdMob
Maximize revenue with in-app ads

AdWords
Drive installs with targeted ad campaigns

Two flavours:

1. *Realtime Database*
   - Just one giant JSON-tree of data
   - ‘Old’
2. *Cloud Firestore*
   - JSON Document collections, like MongoDB, CouchDB, Cassandra, etc.
   - ‘New’ (as of October 2017)
   - Still in beta



Steps:

1. Set up Firebase Project (‘old skool’ Realtime Database version)
2. Set up Angular Project
3. Enhance Angular Project with code
    - CRUD operations
4. Add Authentication
    - Custom Login
    - Social Login
5. Enhance Angular Project
    - Guards, (re-)routing
    - …

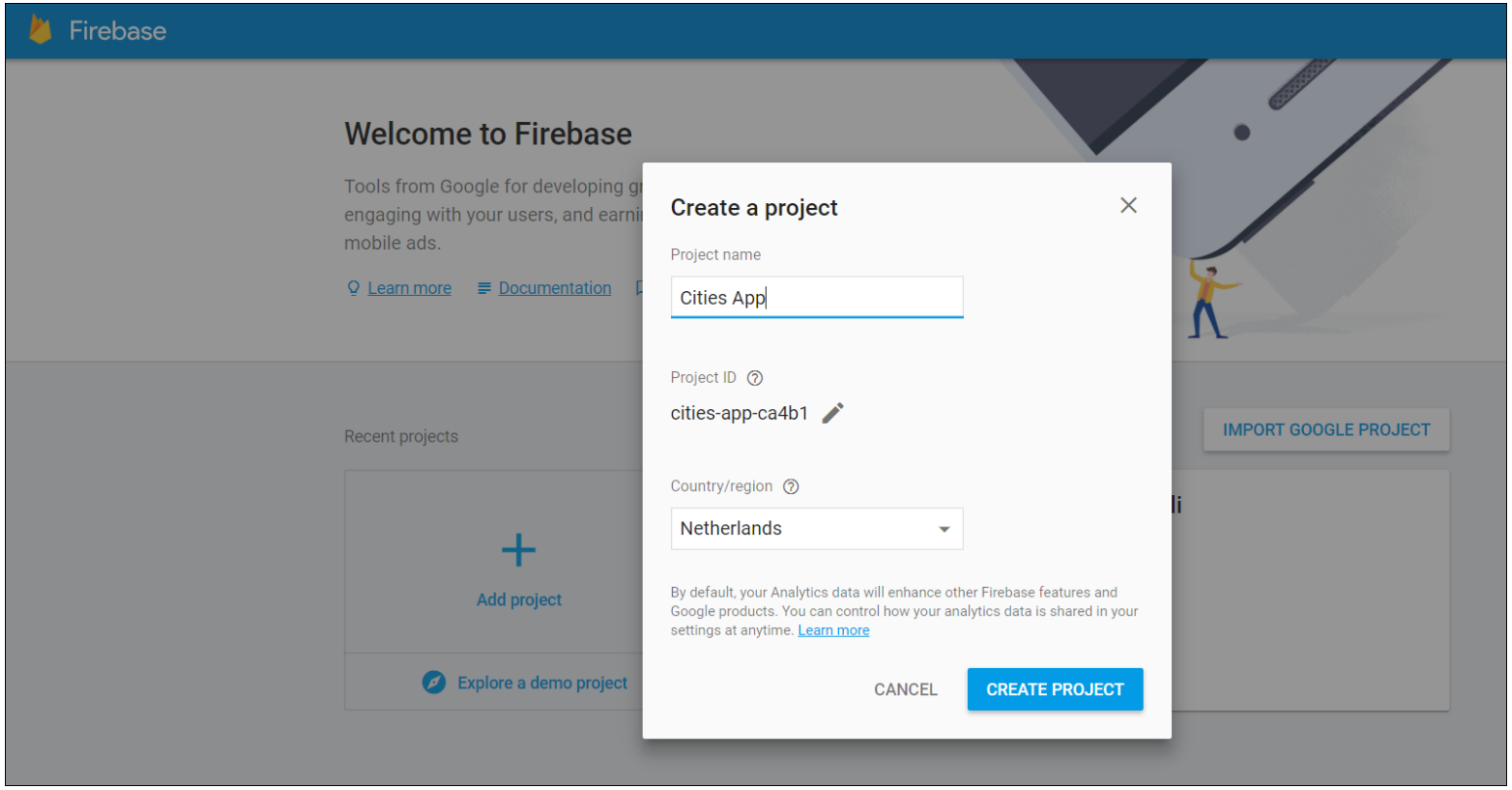
Steps 3-5 can be switched!

Set up the Firebase Project

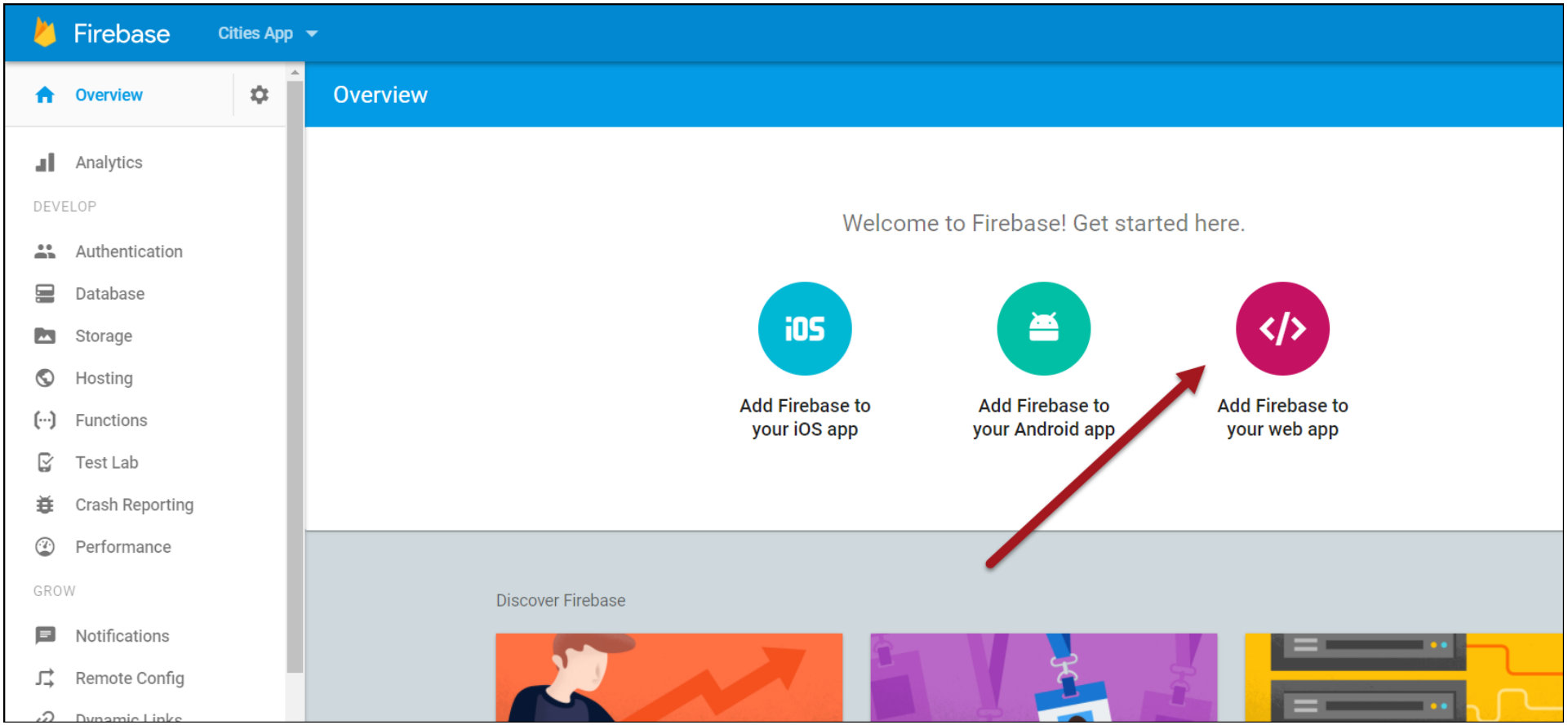
Create account, work with Firebase Console

Set up Firebase Project

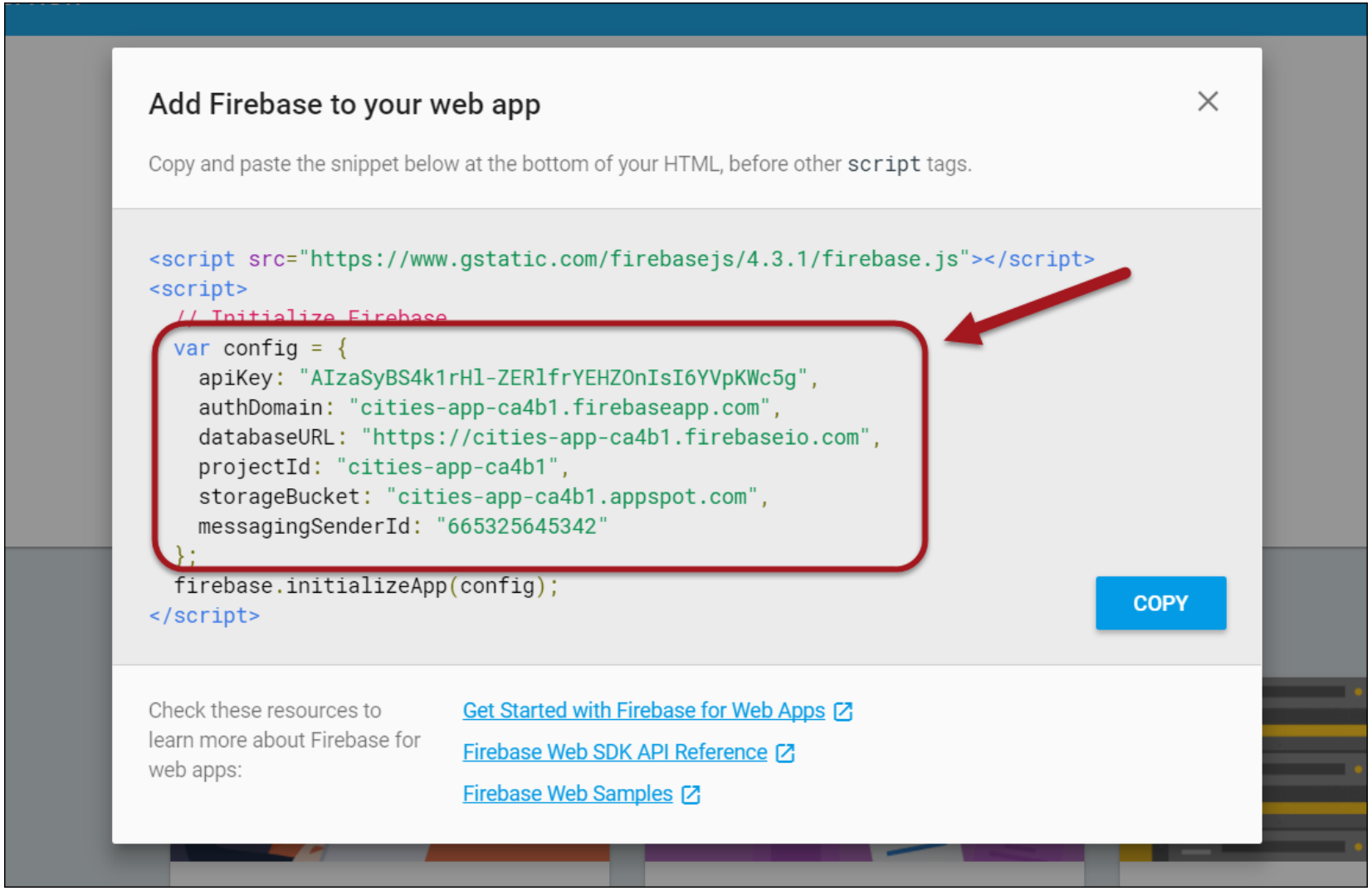
- Go to https://firebase.google.com/
- Log in with your Google Account, click `Go To Console`
- `Create New Project`, and give it a name.



Add Firebase to web app



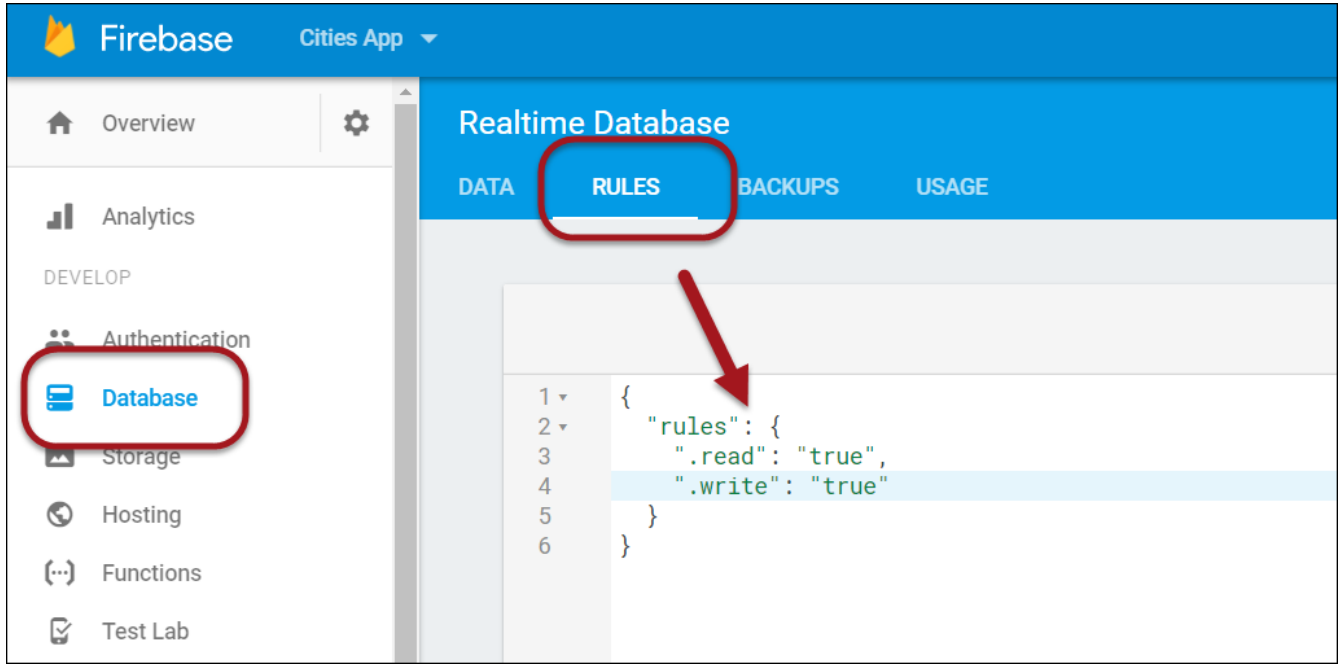
Copy configuration information



Store this in a text file or snippet, for later reference and use in your Angular App

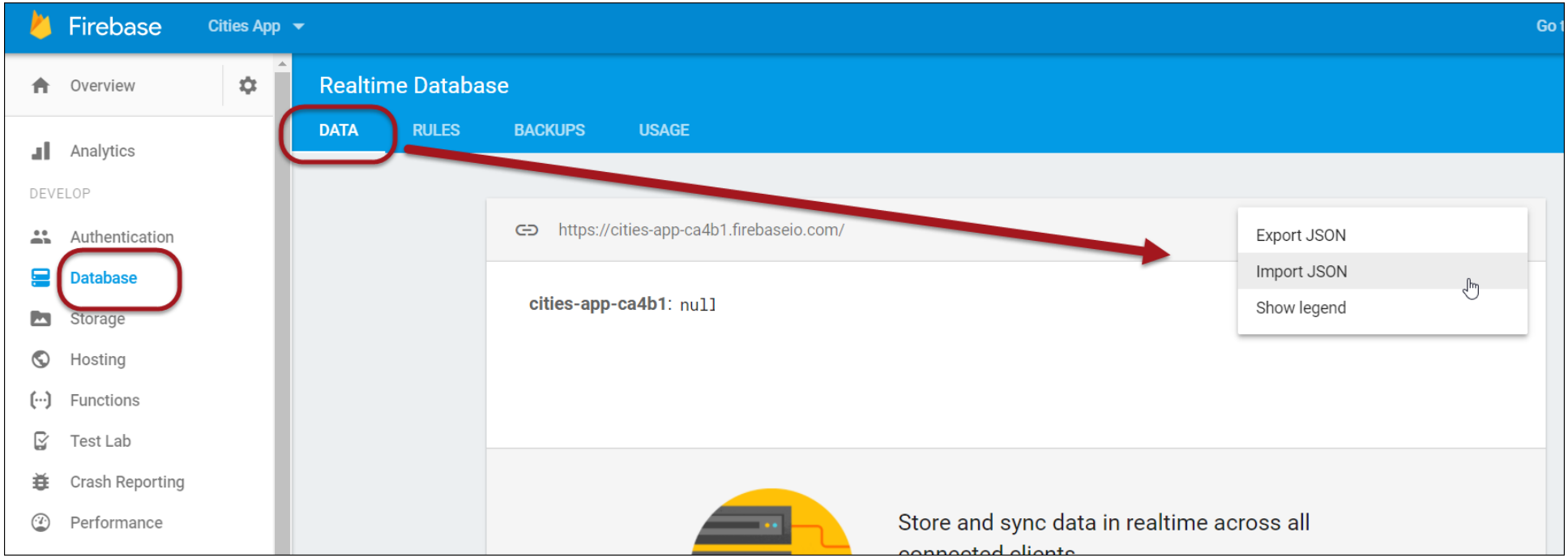
Set database rules

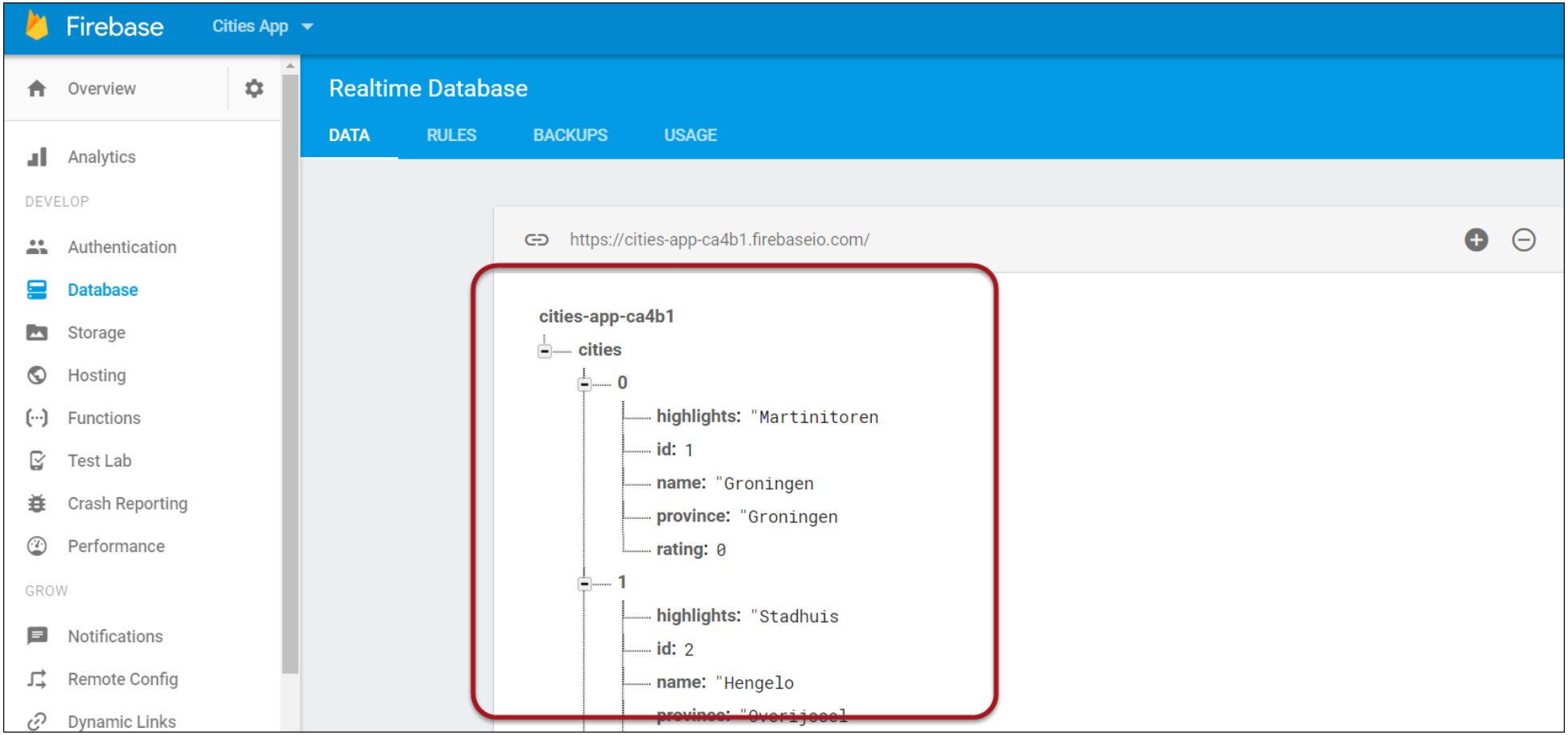
- We're not authenticating users for now – we will later on.
  - DON'T do this in production. This is for development purposes only
- Go to `Database`, `Rules` tab.
- Change `.read`/`.write` values to `true`.



Seed database

- Not absolutely neccesary, we do it to have some data to display
- Firebase can import and export `.json`-files
- Add `cities.json` to the database




Firebase
Cities App
Overview
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Realtime Database
DATA
RULES
BACKUPS
USAGE
https://cities-app-ca4b1.firebaseio.com/
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id: 1
name: "Groningen
province: "Groningen
rating: 0
1
highlights: "Stadhuis
id: 2
name: "Hengelo

Setting up Angular Project

Create new project, add Firebase and AngularFire

AngularFire2

- AngularFire is the Angular Connector to Firebase
- You COULD do this by hand, but using AngularFire it's much easier
- https://github.com/angular/angularfire2

README.md

AngularFire2

The official library for Firebase and Angular 2

build failing chat on gitter

Status: Release candidate

What is AngularFire2?

- **Observable based** - Use the power of rxjs, Angular 2, and Firebase.
- **Realtime bindings** - Synchronize database collections as objects or lists.
- **Authentication** - Monitor authentication state in realtime.

Quick links

Contributing

Stackblitz Template - Remember to set your Firebase configuration in `app/app.module.ts`.

Upgrading to v4.0? Check out our guide.

Install

New Project using the CLI

- Create a new Angular Project using the CLI
  - We also added Bootstrap and Font-awesome for styling
- Inside the project folder, run
- `npm install firebase angularfire2 --save`

Install

```
npm install firebase angularfire2 --save
```

Defining Firebase Settings

- Tell Angular which modules to import and what configuration settings to use
- AngularFire is split up into separate modules:
  - `AngularFireModule`
  - `AngularFireDatbaseModule`
  - `AngularFireAuthModule`
- In `app.module.ts` also add firebase configuration you saved earlier

```
…
// Firebase stuff
import {AngularFireModule} from 'angularfire2';
import {AngularFireDatabaseModule} from 'angularfire2/database';
import {FormsModule} from '@angular/forms';
import {HttpModule} from '@angular/http';

// Initialize Firebase
let firebaseConfig = {
   apiKey           : "AIzaSyBS4k1rHKUY-ERlfrYEHZOnIsI6YVpKWc5g",
   authDomain       : "cities-app-ca4b2.firebaseapp.com",
   databaseURL      : "https://cities-app-ca4b1.firebaseio.com",
   projectId        : "cities-app-ca4b2",
   storageBucket    : "cities-app-ca4b2.appspot.com",
   messagingSenderId: "66532564234612"
};

@NgModule({
   imports     : [
      BrowserModule,
      FormsModule,
      HttpModule,
      AngularFireModule.initializeApp(firebaseConfig),
      AngularFireDatabaseModule
   ],
    …
})
export class AppModule {
}
```

Show data in component

Delete default stuff in `AppComponent.ts`

```
import {Component, OnInit} from '@angular/core';
import {
   AngularFireDatabase, FirebaseListObservable
} from 'angularfire2/database';

@Component({
   selector    : 'app-root',
   templateUrl: './app.component.html’
})
export class AppComponent implements OnInit {
   cities$: FirebaseListObservable<any[]>;

   constructor(private af: AngularFireDatabase) {

   }

   ngOnInit() {
      this.cities$ = this.af.list('/cities');
   }
}
```

Import Firebase stuff

Create Observable

Inject AngularFireDatabase

Get Cities

HTML

```
<!--Fetch a list of cities from Fireabse -->
<h1>List of Cities from Firebase</h1>
<ul class="list-group">
   <li class="list-group-item"
      *ngFor="let city of cities$ | async">
      {{ city.id}} - {{ city.name}}
   </li>
</ul>
```

AngularFireDatabase options

- Fetch data
  - Fetch as Object, import `FirebaseObjectObservable`
  - Fetch as List, import `FirebaseListObservable`
- You get the same data, in different formatting, with different methods

API Summary

The table below highlights some of the common methods on the `FirebaseObjectObservable`.

method	
`set(value: any)`	Replaces the current value in the database with the new value specified as the parameter. This is called a **destructive** u
`update(value: Object)`	Updates the cu called a **non-de** key level are no
`remove()`	Deletes all data

API Summary

The table below highlights some of the common methods on the `FirebaseListObservable`.

method	
`push(value: any)`	Creates a new record on the list, using the Realtime Database's push-ids.
`update(keyRefOrSnap: string)`	Firebase
`remove(key: string?)`	Deletes the item by key. If no parameter is provided, the entire list will be deleted.

AngularFire Documentation

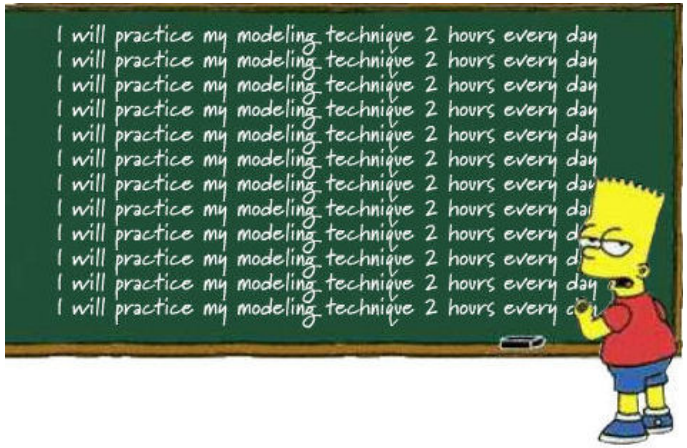
```
  items: FirebaseListObservable<any[]>;
  constructor(db: AngularFireDatabase) {
    this.items = db.list('/items');
  }
}
```

Developer Guide

If you want to get started quickly on building with AngularFire2, check out our 5 step developer guide that will teach you everything you need to know to be productive with AngularFire2.

1. Installation & Setup
2. Retrieving data as objects - FirebaseObjectObservable
3. Retrieving data as lists - FirebaseListObservable
4. Querying lists
5. User Authentication - FirebaseAuthentication
6. Using AngularFire2 with Ionic 2
7. Using AngularFire2 with Ionic 3 and Angular 4
8. Deploying AngularFire2 to FirebaseHosting

Sparse, but it will get you started: https://github.com/angular/angularfire2

Workshop

- Get yourself a Firebase account and create project in the console
- Don't forget to set `.read/.write` values to `true` (for the moment)
- Create a new, empty Angular CLI - project
- Add Firebase + AngularFire to it
- Get data from database and show it in the UI
- Experiment with different options for `FirebaseObjectObservable` and `FirebaseListObservable` (for example: querying a List)
- https://github.com/angular/angularfire2
- Example: `/500-Firebase`